

## IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

### PROJECT 29

# Add Private Subnets and NAT Gateway Egress

Amazon VPC | Private Subnets | NAT Gateway | Route Tables | Elastic IP

|              |                  |
|--------------|------------------|
| AUTHOR       | Michael Salazar  |
| ENVIRONMENT  | Personal AWS Lab |
| VERSION      | v1.0.0           |
| RELEASE DATE | July 21, 2026    |
| STATUS       | Complete         |

## Executive Summary

Extended the custom **CloudSLAW** VPC by creating two private subnets in **us-west-2a** and **us-west-2b**, creating a **NAT Gateway** in the public subnet **slaw-public-1**, creating a dedicated **slaw-private-route** route table, routing private-subnet Internet-bound traffic to the NAT Gateway, and associating **slaw-private-1** and **slaw-private-2** with the private route table. The lab was then cleaned up by deleting the NAT Gateway, releasing the Elastic IP address, and deleting the VPC to avoid NAT Gateway and Elastic IP costs.

### COST CONTROL

This lab intentionally created a NAT Gateway and then removed it. NAT Gateways and Elastic IP addresses can generate charges even when no workload traffic is flowing, so cleanup is a required part of the lab.

## Business Problem

Project 28 created public subnets that route directly to an Internet Gateway. That is useful for public-facing resources, but many workloads need outbound Internet access for updates or AWS API calls without being directly reachable from the Internet. A NAT Gateway supports that pattern by allowing outbound traffic from private subnets while preventing unsolicited inbound Internet traffic from directly reaching those private subnet resources.

## Objectives

- Create two new private subnets in the existing CloudSLAW VPC.
- Use private subnet CIDR ranges 10.0.3.0/24 and 10.0.4.0/24.
- Place private subnets across us-west-2a and us-west-2b.
- Create a NAT Gateway in the public subnet slaw-public-1.
- Allocate an Elastic IP address for the public NAT Gateway.
- Create a dedicated private route table named slaw-private-route.
- Add a default route so 0.0.0.0/0 traffic from private subnets targets the NAT Gateway.
- Associate slaw-private-1 and slaw-private-2 with the private route table.
- Clean up the lab by deleting the NAT Gateway, releasing the Elastic IP address, and deleting the VPC.

## Environment

| Cloud Provider  | Amazon Web Services (AWS)       |
|-----------------|---------------------------------|
| Primary Service | Amazon VPC                      |
| Account Context | TestAccount1 / workload account |
| Region          | us-west-2 / Oregon              |

|                         |                                                                                                                                                                                                                                                               |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VPC                     | CloudSLAW                                                                                                                                                                                                                                                     |
| VPC CIDR                | 10.0.0.0/16                                                                                                                                                                                                                                                   |
| Existing Public Subnets | slaw-public-1: 10.0.1.0/24, slaw-public-2: 10.0.2.0/24                                                                                                                                                                                                        |
| New Private Subnets     | slaw-private-1: 10.0.3.0/24, slaw-private-2: 10.0.4.0/24                                                                                                                                                                                                      |
| NAT Gateway Placement   | slaw-public-1                                                                                                                                                                                                                                                 |
| Private Route Table     | slaw-private-route                                                                                                                                                                                                                                            |
| Private Default Route   | 0.0.0.0/0 -> NAT Gateway                                                                                                                                                                                                                                      |
| Cleanup                 | Deleted NAT Gateway, released Elastic IP, and deleted VPC                                                                                                                                                                                                     |
| Evidence Limitation     | Evidence shows route-table update, private subnet associations, and subnet inventory. NAT Gateway deletion, Elastic IP release, and VPC deletion are documented from the implementation statement and lab cleanup requirement, not from separate screenshots. |

## Architecture

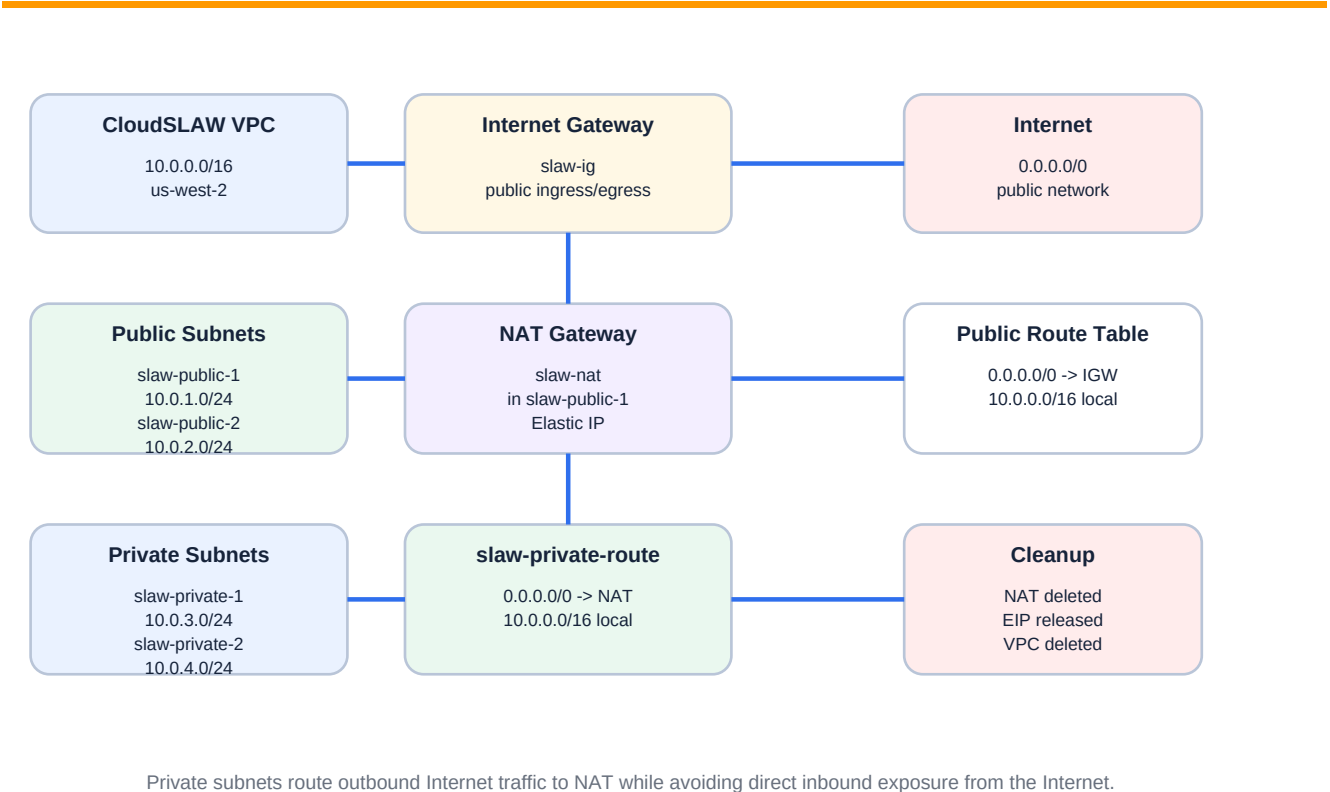


Figure 1 - Public and private subnet design with NAT Gateway outbound egress.

## Implementation Summary

1. Started from the custom CloudSLAW VPC created in Project 28.

2. Created slaw-private-1 in us-west-2a with CIDR 10.0.3.0/24.
3. Created slaw-private-2 in us-west-2b with CIDR 10.0.4.0/24.
4. Created a public NAT Gateway in slaw-public-1 and allocated an Elastic IP address.
5. Created a new route table named slaw-private-route.
6. Edited the private route table to add destination 0.0.0.0/0 with the NAT Gateway as the target.
7. Associated slaw-private-1 and slaw-private-2 with slaw-private-route.
8. Validated that public subnets remained associated with the main route table while private subnets were explicitly associated with the private route table.
9. Deleted the NAT Gateway, released the assigned Elastic IP address, and deleted the VPC to avoid ongoing lab charges.

## Routing Design

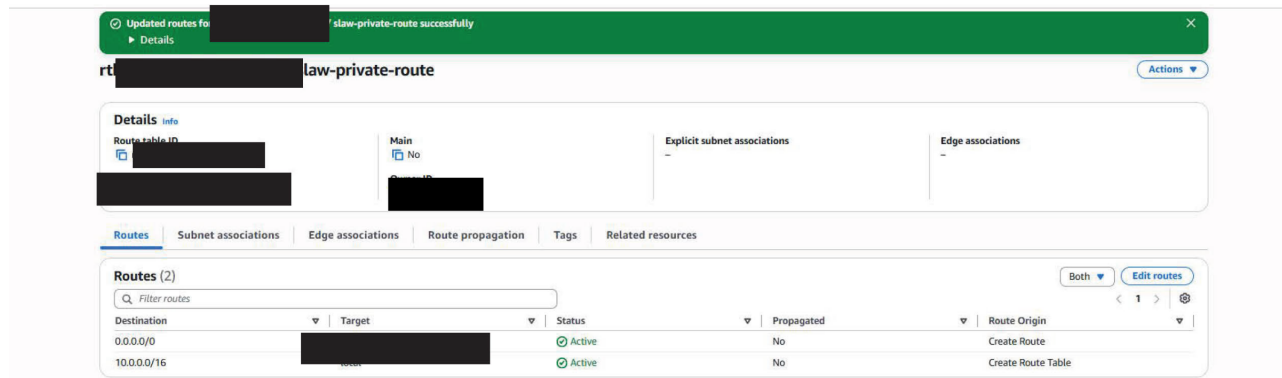
| Public                | 0.0.0.0/0 -> Internet Gateway | Can support direct Internet access when resources have public IPs and security groups allow traffic. |
|-----------------------|-------------------------------|------------------------------------------------------------------------------------------------------|
| Private               | 0.0.0.0/0 -> NAT Gateway      | Allows outbound Internet access while avoiding direct inbound Internet routing.                      |
| Local                 | 10.0.0.0/16 -> local          | Allows communication inside the VPC address space.                                                   |
| NAT Gateway placement | Public subnet slaw-public-1   | NAT must be in a public subnet so it can reach the Internet Gateway.                                 |
| Elastic IP            | Allocated to NAT Gateway      | Provides the Internet-routable address used by the NAT Gateway.                                      |

## Evidence

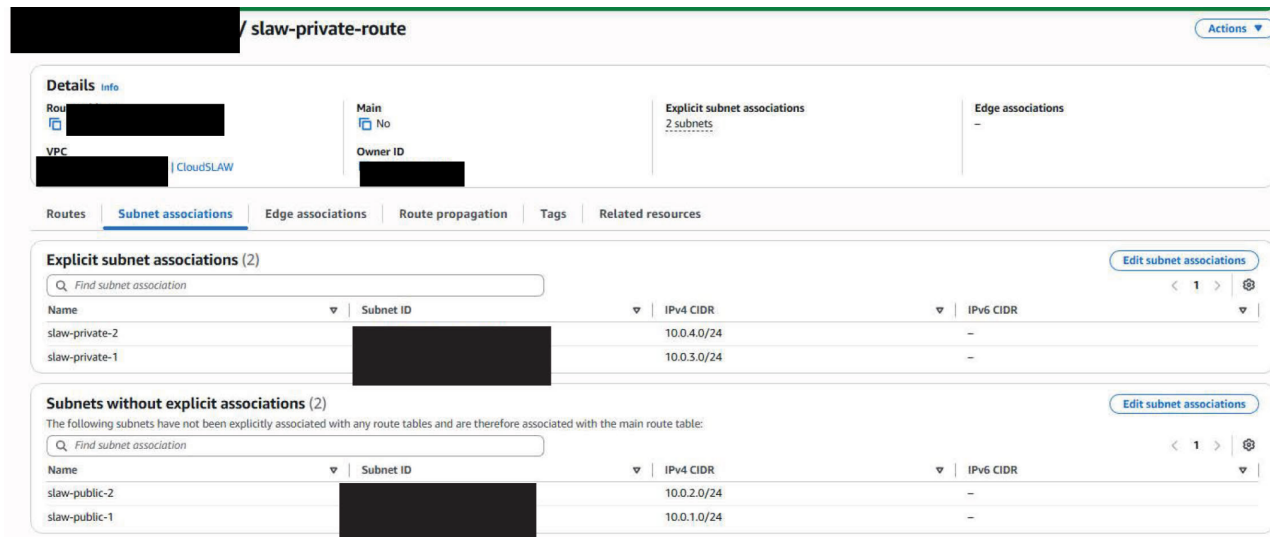
**Figure 2 - Four Subnets Created in the CloudSLAW VPC**

| Name           | Subnet ID | State     | VPC | Block Public... | IPv4 CIDR   | IPv6 CIDR | IPv6 CIDR association ID | Availi |
|----------------|-----------|-----------|-----|-----------------|-------------|-----------|--------------------------|--------|
| slaw-private-2 |           | Available |     | Off             | 10.0.4.0/24 | -         | -                        | 251    |
| slaw-public-2  |           | Available |     | Off             | 10.0.2.0/24 | -         | -                        | 251    |
| slaw-public-1  |           | Available |     | Off             | 10.0.1.0/24 | -         | -                        | 251    |
| slaw-private-1 |           | Available |     | Off             | 10.0.3.0/24 | -         | -                        | 251    |

The subnet inventory evidence shows four subnets in the CloudSLAW VPC: slaw-public-1 with 10.0.1.0/24, slaw-public-2 with 10.0.2.0/24, slaw-private-1 with 10.0.3.0/24, and slaw-private-2 with 10.0.4.0/24. Each subnet shows an Available state and 251 available IPv4 addresses.

**Figure 3 - Private Route Table Updated for NAT Egress**

The route-table evidence shows *slaw-private-route* updated successfully. The route table contains two active routes: 10.0.0.0/16 for local VPC routing and 0.0.0.0/0 targeting the NAT Gateway.

**Figure 4 - Private Subnets Associated with slaw-private-route**

The subnet-association evidence shows two explicit subnet associations on *slaw-private-route*: *slaw-private-1* with 10.0.3.0/24 and *slaw-private-2* with 10.0.4.0/24. The public subnets remain without explicit association and therefore use the main route table.

## Security Analysis

This design separates public and private network paths. Public subnets use the Internet Gateway route, while private subnets use a NAT Gateway for outbound-only Internet access.

The key difference is the route target for 0.0.0.0/0. An Internet Gateway route makes a subnet public; a NAT Gateway route supports private-subnet outbound connectivity.

The NAT Gateway is placed in *slaw-public-1* because the NAT Gateway itself needs a public path through the Internet Gateway.

Using a separate *slaw-private-route* table creates explicit control over which subnets are treated as private.

The lab cleanup is important because NAT Gateway and Elastic IP resources can create ongoing cost. The cleanup step is part of the control, not an afterthought.

# Lessons Learned

Private subnets are not private because of the subnet name. They are private because their route table does not send 0.0.0.0/0 directly to an Internet Gateway.

NAT provides outbound access for private resources but does not create a direct inbound route from the Internet to those resources.

Route table associations define which subnets receive a routing policy.

A NAT Gateway requires an Elastic IP address when configured for public outbound Internet access.

Networking labs should include cleanup steps because network resources such as NAT Gateways can create charges.

# Production Improvements

- Use one NAT Gateway per Availability Zone for production high availability.
- Route each private subnet to the NAT Gateway in the same Availability Zone to avoid cross-AZ dependency and data-transfer cost.
- Add VPC Flow Logs to monitor traffic from public and private subnets.
- Consider VPC endpoints for AWS services such as S3 and Systems Manager to reduce NAT Gateway traffic and cost.
- Add explicit route-table names, subnet tags, and environment tags.
- Add security groups and network ACL review before deploying workloads.
- Use infrastructure as code to recreate the VPC consistently after lab cleanup.

# Skills Demonstrated

| Amazon VPC              | NAT Gateway     | Elastic IP          |
|-------------------------|-----------------|---------------------|
| Private subnets         | Route tables    | Subnet associations |
| Internet Gateway design | Outbound egress | Lab cleanup         |

# Resume Bullet

RESUME

Implemented private subnet outbound egress in AWS by creating private subnets, deploying a NAT Gateway in a public subnet, configuring a dedicated private route table with 0.0.0.0/0 to NAT, associating private subnets, and cleaning up NAT, Elastic IP, and VPC resources to control costs.

## STAR Interview Story

---

**Situation:** The custom VPC had public subnets with Internet Gateway routing, but still needed private subnets that could reach the Internet without being directly exposed.

**Task:** Add private subnet routing through NAT and then clean up the cost-generating resources after the lab.

**Action:** I created two private subnets, created a NAT Gateway in slaw-public-1 with an Elastic IP, built slaw-private-route, routed 0.0.0.0/0 to NAT, associated both private subnets, then deleted the NAT Gateway, released the Elastic IP, and deleted the VPC.

**Result:** The lab demonstrated the difference between public and private subnet routing and established a cost-aware private egress design pattern.

## Version History

---

| v1.0.0 | July 21, 2026 | Initial documented implementation. Created private subnets, NAT Gateway, private route table, 0.0.0.0/0 NAT route, private subnet associations, cleanup notes for NAT Gateway, Elastic IP, and VPC deletion, and embedded provided evidence using portfolio documentation standard v1.0.1. |
|--------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## References

---

- Cloud Security Lab a Week (S.L.A.W.) - NAT Your Way to Privacy (and Maybe Poverty).
- User-provided route-table update screenshot.
- User-provided subnet-association screenshot.
- User-provided subnet inventory screenshot.
- Project 28 - Build a Custom VPC with Public Subnets and Internet Gateway Routing.